



Quantum Face Recognition with Multi-Gate Quantum Convolutional Neural Network

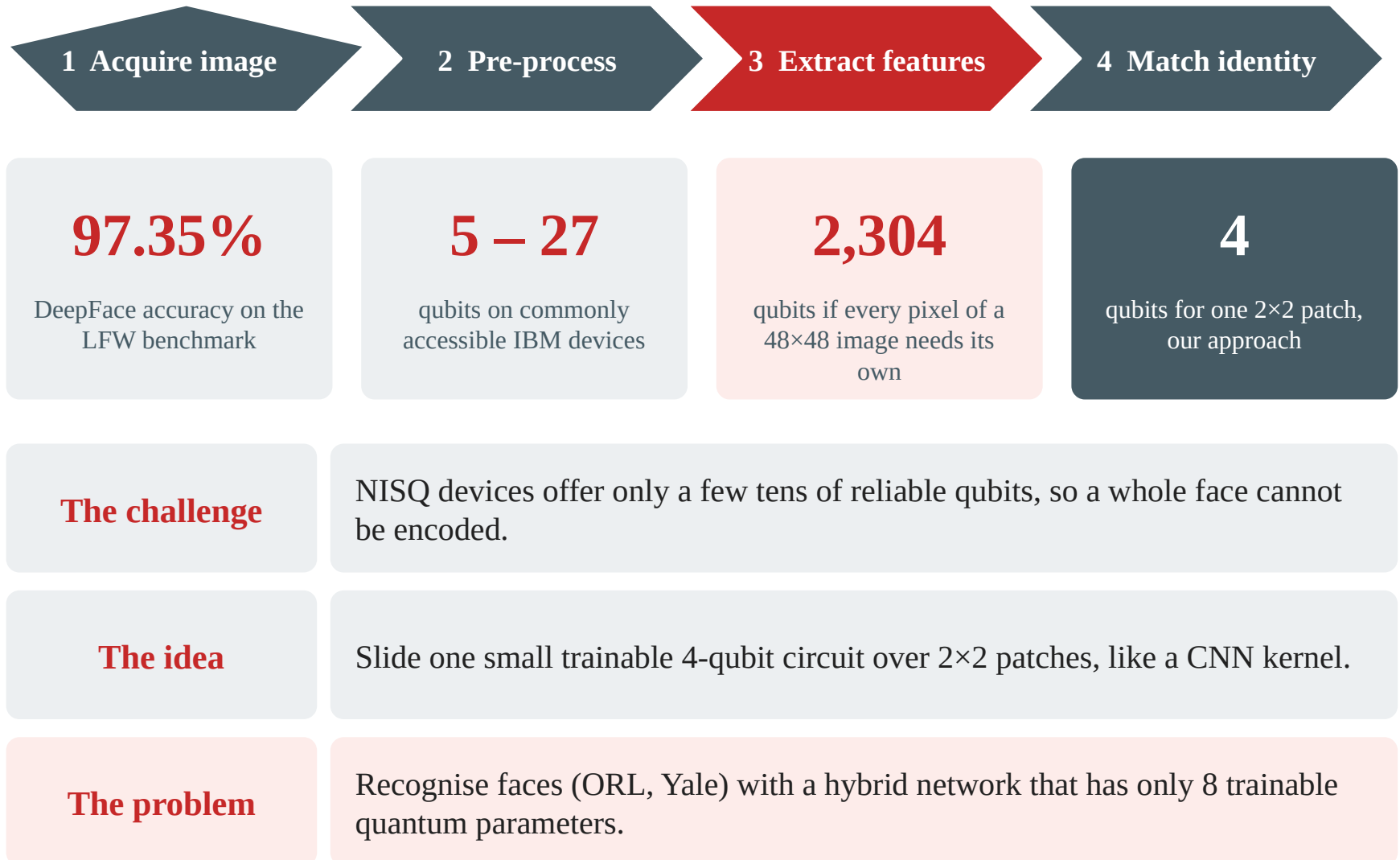
B. Tech. Project Mid-Semester Evaluation

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Introduction & Problem Statement



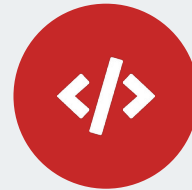
Objectives



1

Study quantum ML

Angle encoding, variational circuits, quanvolution



2

Implement the MG-QCNN

PennyLane + PyTorch, with an analytical model of the filter



3

Evaluate on ORL and Yale

10 random 70:30 splits, 1000 measurement shots



4

Analyse the design

Gates, entanglement, filter size, shot noise, hardware cost

Literature Review

Classical face recognition

- Gabor + KNN/SVM gave way to deep CNNs
- High accuracy, heavy computation
- Compute demand outgrows hardware

Variational circuits

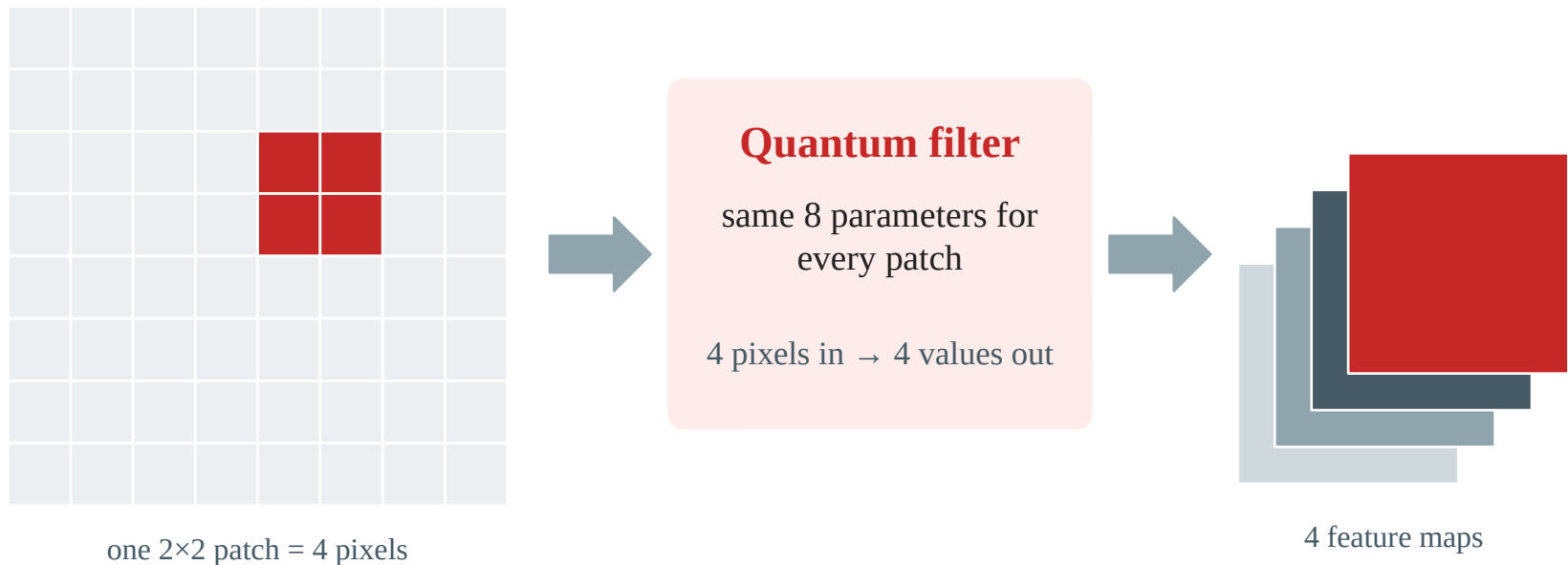
- Rotation angles act as weights
- Trained by gradient descent
- Barren plateaus favour small, fixed circuits

Quantum convolution

- Henderson: random, fixed filter
- Mattern: trainable but random
- Oh: fixed, one measured qubit
- MG-QCNN: fixed, trainable, all 4 qubits measured

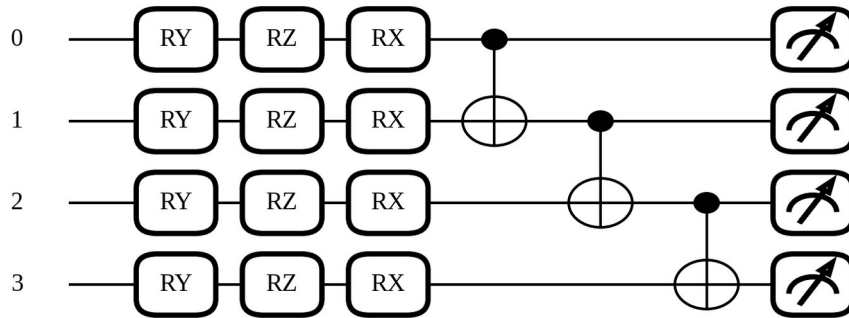
Research gap: small datasets, noise-free simulation and no hardware test in the reference work. We add an independent implementation, an analytical model of the filter and a cost and noise analysis.

System Architecture



$$24 \times 24 = 576 \text{ patches} \times 4 \text{ outputs} = 2,304 \text{ features}$$

Quantum Filter: Circuit & Closed Form



Encode

RY(x) per pixel

Train

RZ, RX: 8 θ

Entangle + measure

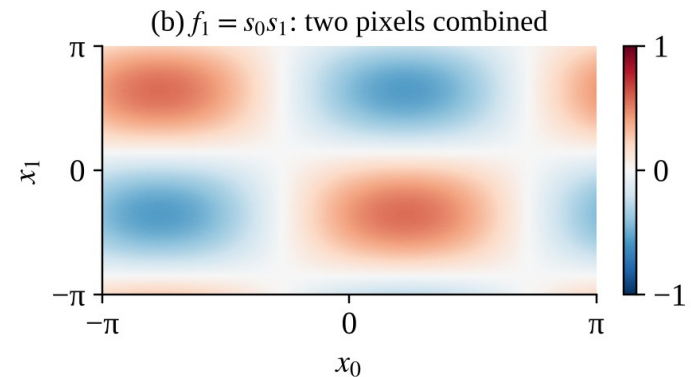
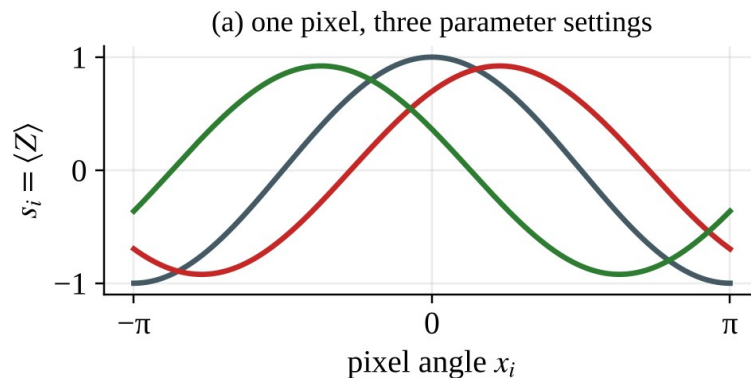
CNOT chain, $\langle Z \rangle$

Closed form

$$s_i = \cos \theta_{2i+1} \cos x_i + \sin \theta_{2i+1} \sin \theta_{2i} \sin x_i$$

$$f_k = s_0 s_1 \cdots s_k$$

f_0 sees 1 pixel, f_3 sees all 4 pixels. Checked against the full simulation: error 7×10^{-16}



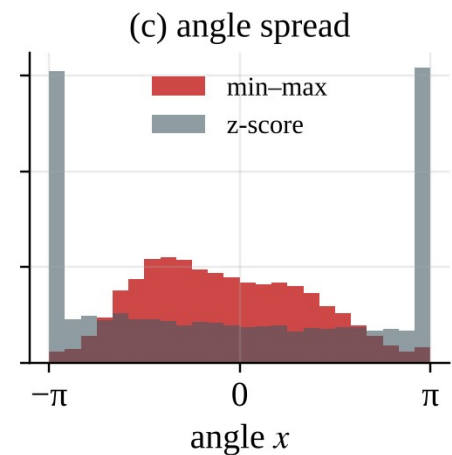
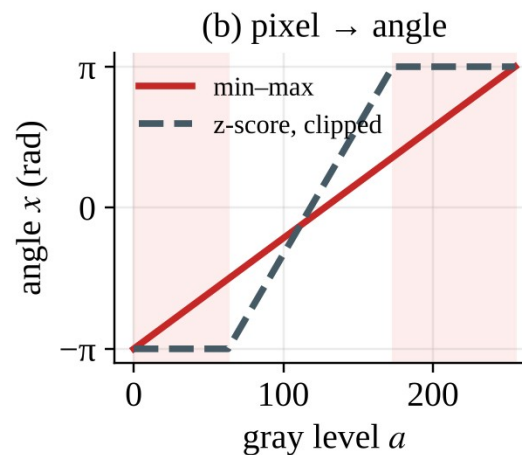
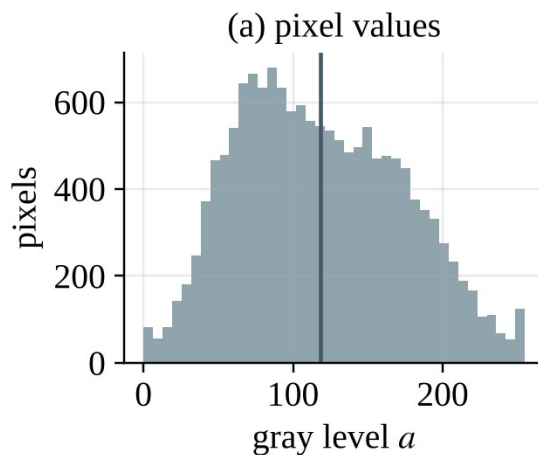
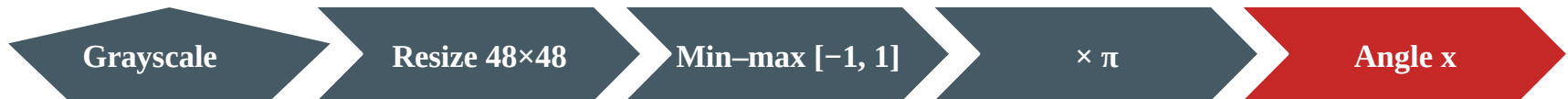
Data & Pre-processing

ORL 400 images · 40 subjects

92×112 px · 10 per subject · split 280 / 120

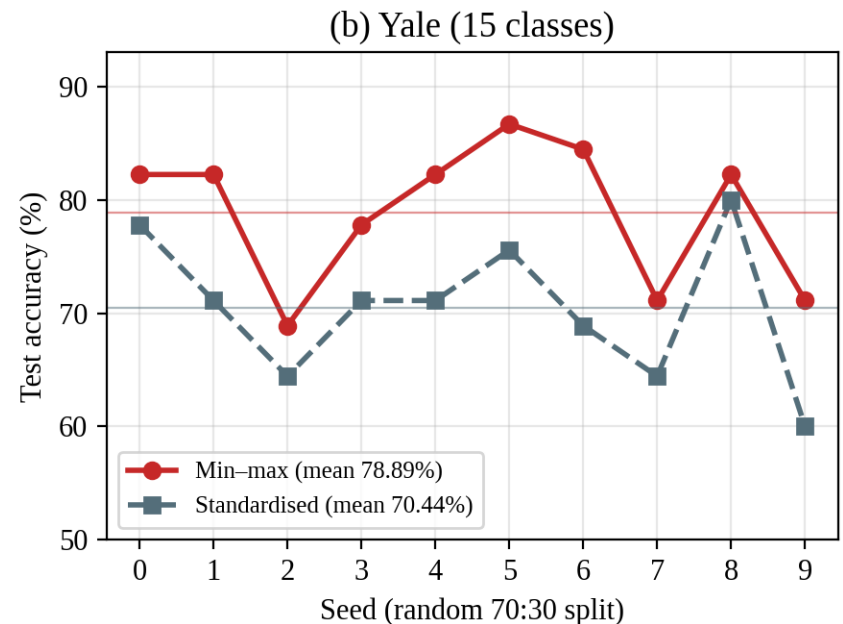
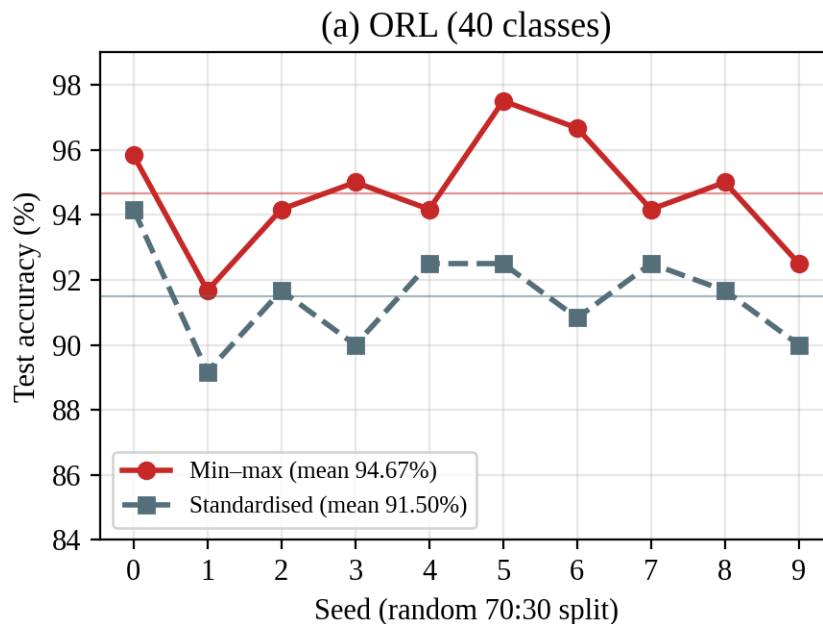
Yale 165 images · 15 subjects

320×243 px · 11 per subject · split 120 / 45



Illustrative (synthetic gray levels): clipped z-scores pile about a third of the pixels at $\pm\pi$; min-max keeps the full angle range.

Effect of Pixel Scaling



10 / 10

seeds won by min-max, both datasets

+3.17 pts

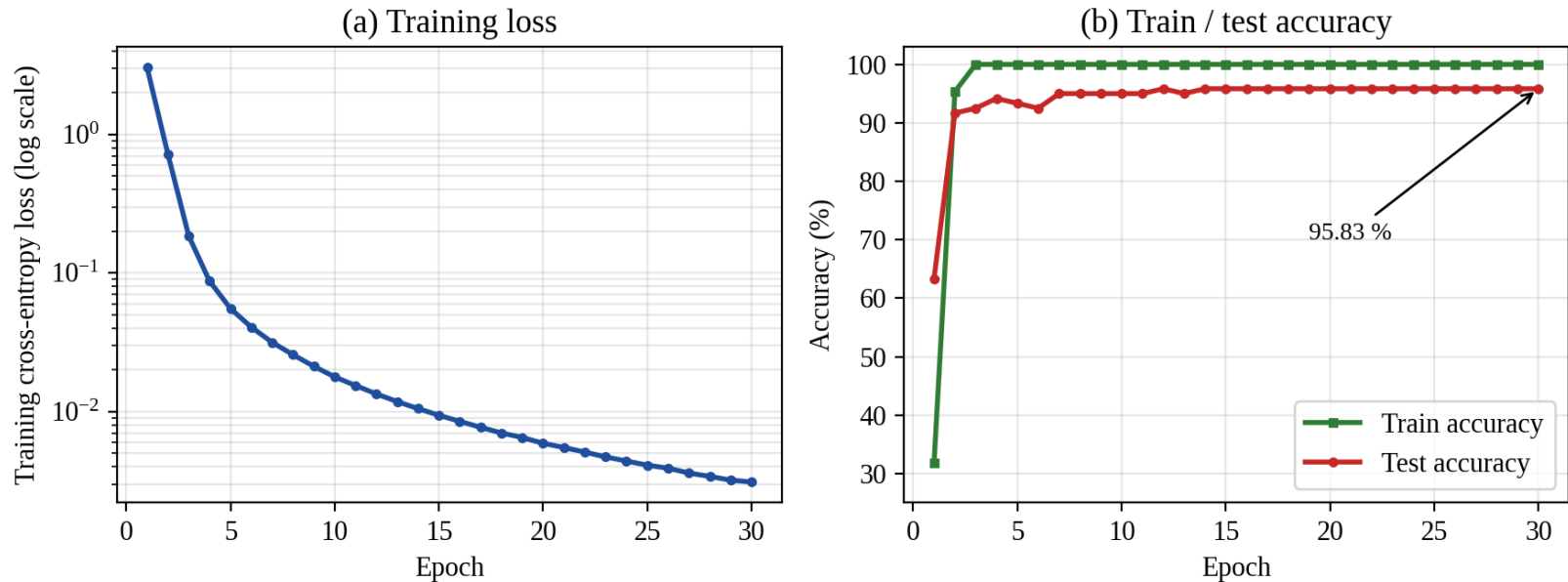
ORL mean: 94.67 vs 91.50

+8.44 pts

Yale mean (original set-up): 78.89 vs
70.44

Likely reason (untested): clipped z-scores saturate bright and dark pixels at $\pm\pi$.

Training Behaviour on ORL



3.03 → 0.0031

training loss, epoch 1 to 30

100%

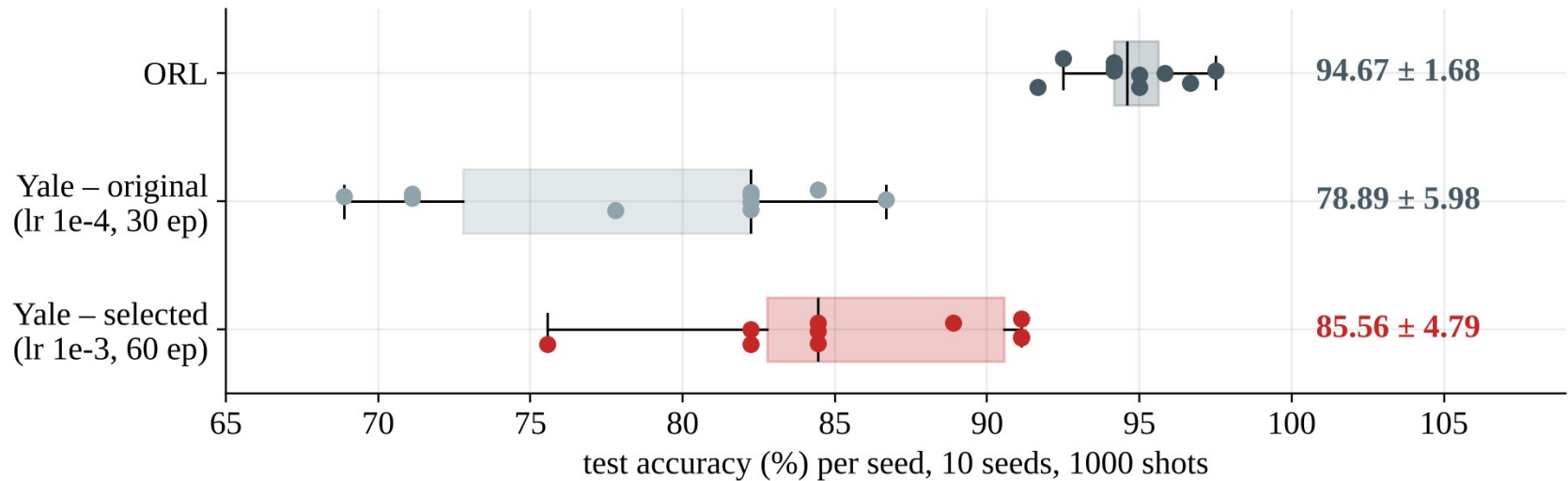
train accuracy from epoch 3

95.83%

test accuracy from epoch 14 (115 of 120)

Some over-fitting: 4.17-point train–test gap, with 92,200 classical weights on 280 images.

Ten-Seed Evaluation



94.67%

ORL mean · max 97.50 · std 1.68 ·
120 test images

85.56%

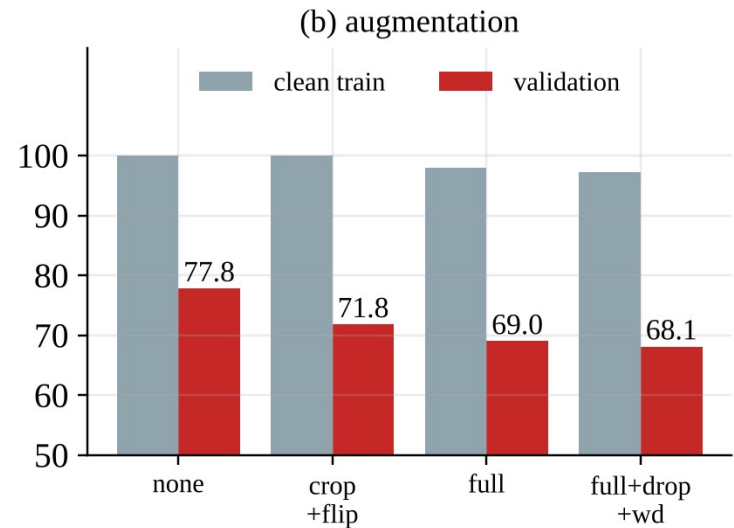
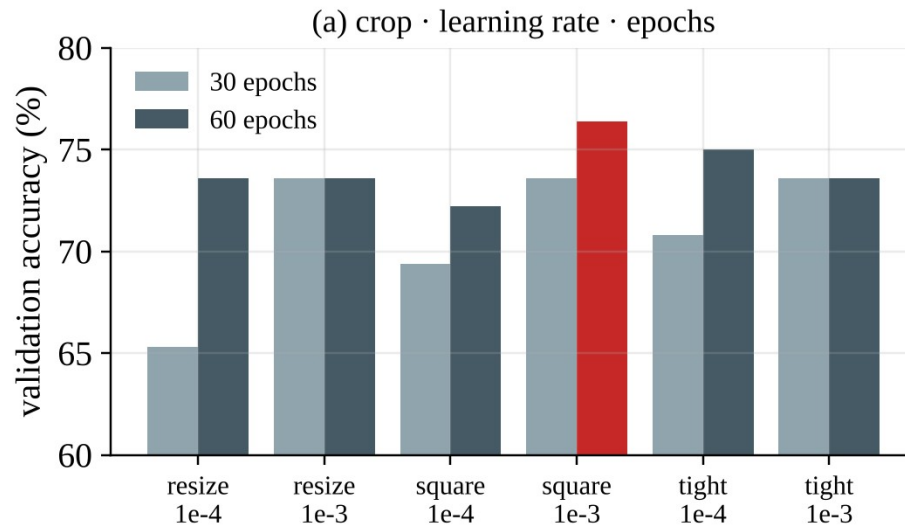
Yale mean · max 91.11 · std 4.79 · 45
test images

+6.7 pts

Yale after centre crop, lr 1e-3, 60
epochs (from 78.89%)

Why Yale varies more: one test image moves accuracy by 2.22 points (45 images) vs 0.83 on ORL (120).

Yale: Tuning and Augmentation



76.4%

best validation: square crop, lr 1e-3,
60 epochs

77.8 → 68.1%

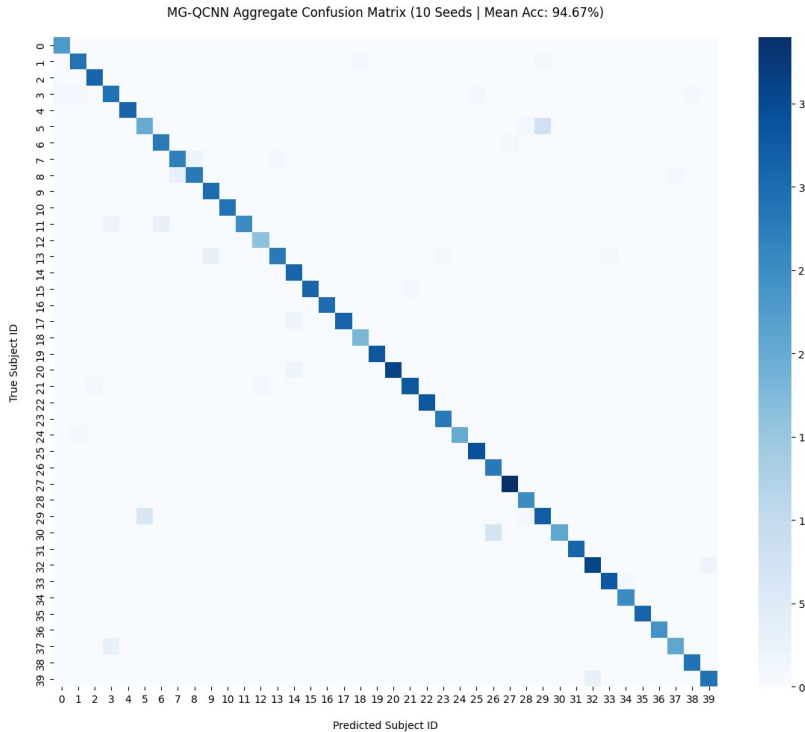
validation accuracy as augmentation
increases

22 → 29 pts

train-validation gap grows

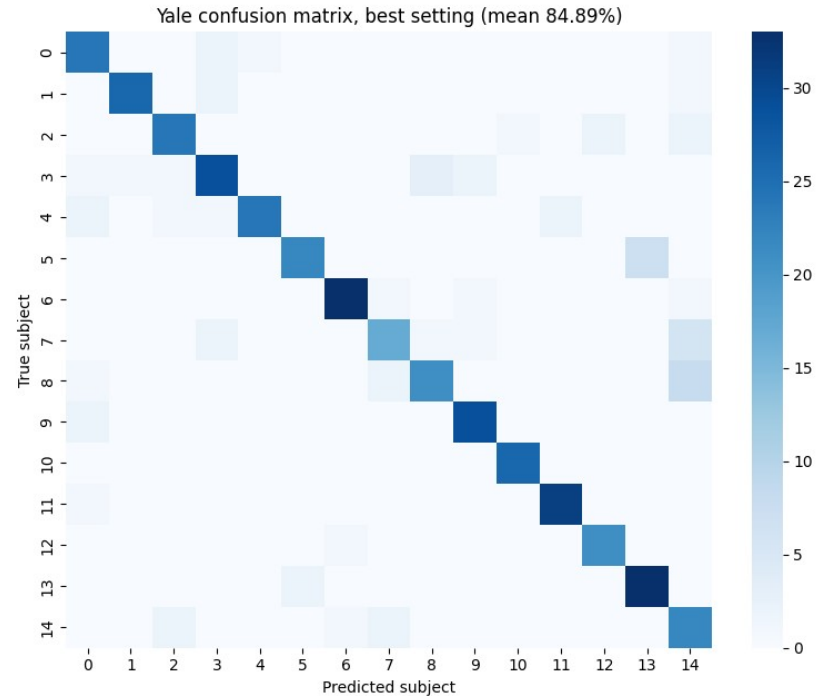
Augmentation did not help: the model still fits 120 training images almost perfectly. The cause of the overfitting is not yet isolated.

Where the Errors Are



ORL · 94.67% · 1,200 predictions

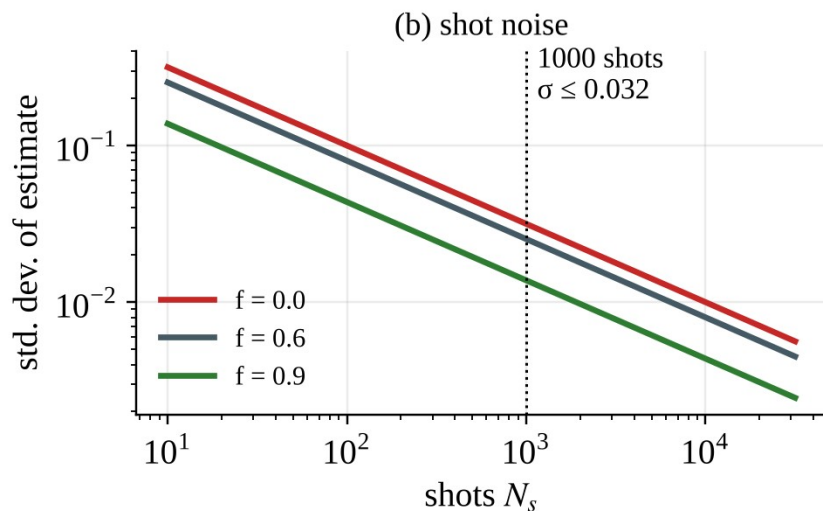
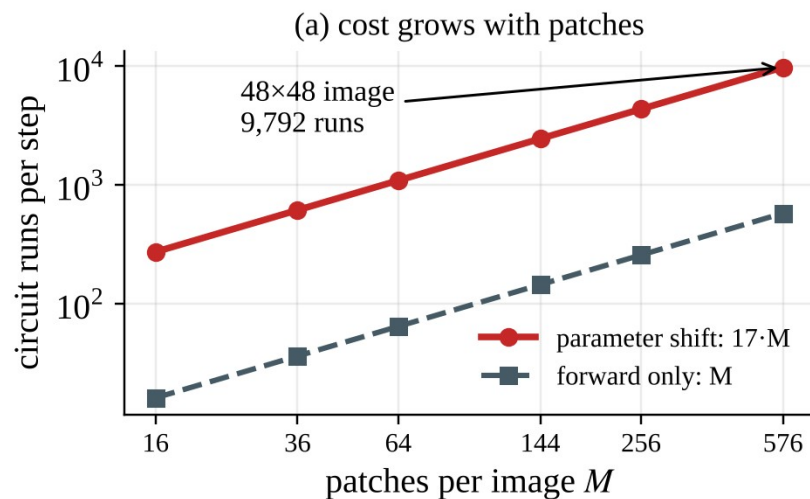
ORL: strong diagonal; errors are scattered single confusions, no subject fails systematically.



Yale (tuning run) · 84.89% · 450 predictions

Yale: more off-diagonal confusion; subjects 7 and 8 are often read as 14, subject 5 as 13.

Cost of Training on Hardware



$$\partial f / \partial \theta = \frac{1}{2} [f(\theta + \pi/2) - f(\theta - \pi/2)]$$

two extra runs per parameter on hardware; simulator
backprop gives the same gradient in one pass

17

runs per patch ($1 + 2 \times 8$)

9,792

runs per image per step

8.2×10^7

runs for one ORL training
run

2.2×10^{-15}

backprop vs shift gradient
mismatch

Conclusion, Limitations & Future Scope



Achieved

- MG-QCNN built in PennyLane + PyTorch
- Closed-form filter verified
- ORL 94.67%, Yale 85.56%
- Min-max beats Gaussian on all seeds



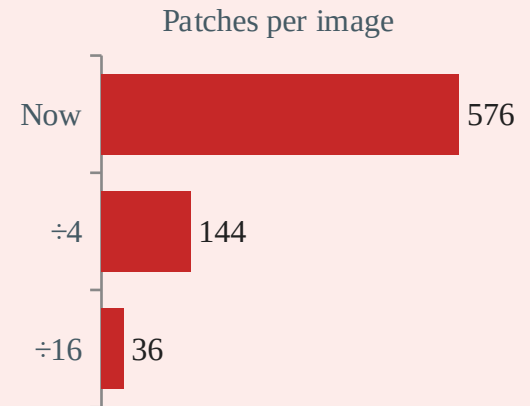
Limitations

- Noise-free simulation, not hardware
- Yale over-fits: 100% train vs about 85% test
- No ablation or noise study yet



Future scope

Add a feature-extraction stage before the quantum layer to cut the patches needing circuit runs



8 quantum parameters · ORL 94.67% · Yale 85.56%

Fewer patches mean 4× to 16× fewer circuit runs and make real parameter-shift training practical.

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